CYBER SECURITY INTERNSHIP REPORT

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Domain: Ethical Hacking
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Projects and Tasks (Intermediate Level)

Assessment 3: Anonymous Communication Using Tor Network

1. Introduction

The main objective of this task is to understand how anonymous communication works using the Tor network and to implement IP rotation for maintaining operational security.

In cybersecurity, it is important to hide the real identity of the system while performing analysis or testing. In this task, I used Tor along with Proxychains and browser configuration to route traffic anonymously. I also verified anonymity using IP and DNS leak testing tools.

2. Understanding Tor Network

Tor (The Onion Router) is a privacy-focused network that hides user identity by routing traffic through multiple nodes.

How Tor Works:

- Data is encrypted in multiple layers
- It passes through:
    - Entry Node
    - Relay Node
    - Exit Node
- The website only sees the **Exit Node IP**, not the real IP

3. Lab Environment

- Operating System: Kali Linux
- Tools Used:
    - Tor Service
    - Proxychains
    - Firefox Browser
    - Curl
- Internet Connection: Required

4. Step-by-Step Implementation

4.1 Installing Tor

```
sudo apt update
sudo apt install tor -y
```

4.2 Starting Tor Service

```
sudo service tor@default start
```

Tor was successfully started and initialized.

4.3 Verifying Tor Connection

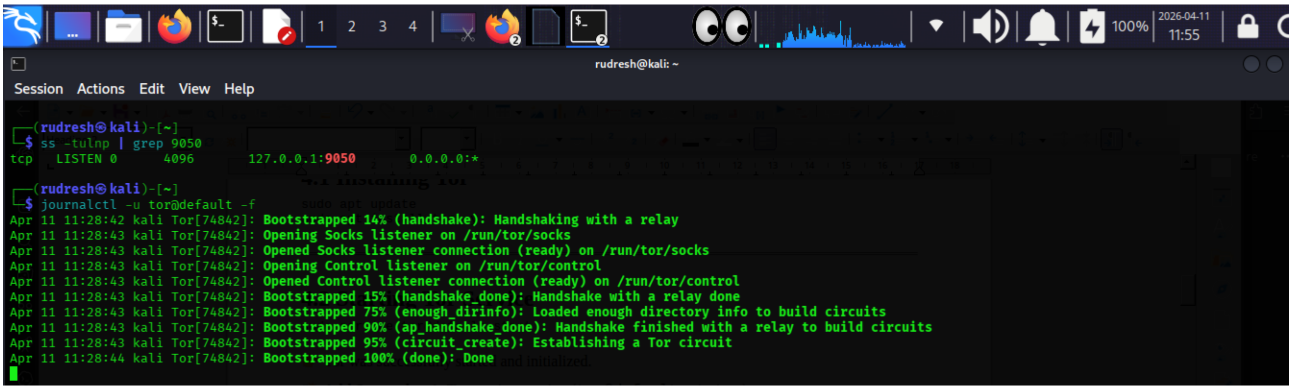
journalctl -u tor@default -f

Tor logs showed:

```
Bootstrapped 100% (done)
```

This confirms Tor network is fully connected.

Output:



4.4 Checking Tor Port

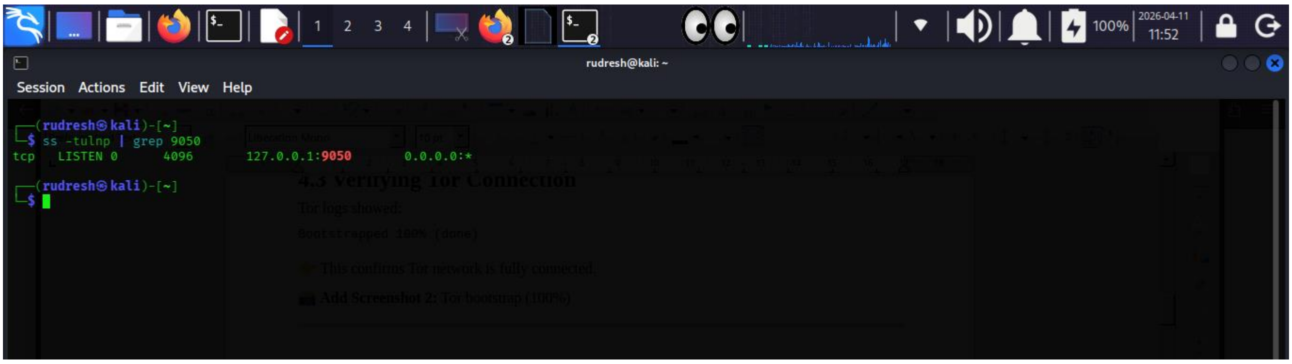
```
ss -tulnp | grep 9050
```

Output:

```
127.0.0.1:9050 LISTEN
```

This confirms SOCKS proxy is active.

Result:



4.5 Installing Proxychains

```
sudo apt install proxychains -y
```

4.6 Configuring Proxychains

Edited:

```
sudo nano /etc/proxychains.conf
```

Changes made:

- Enabled `dynamic_chain`
- Enabled `proxy_dns`

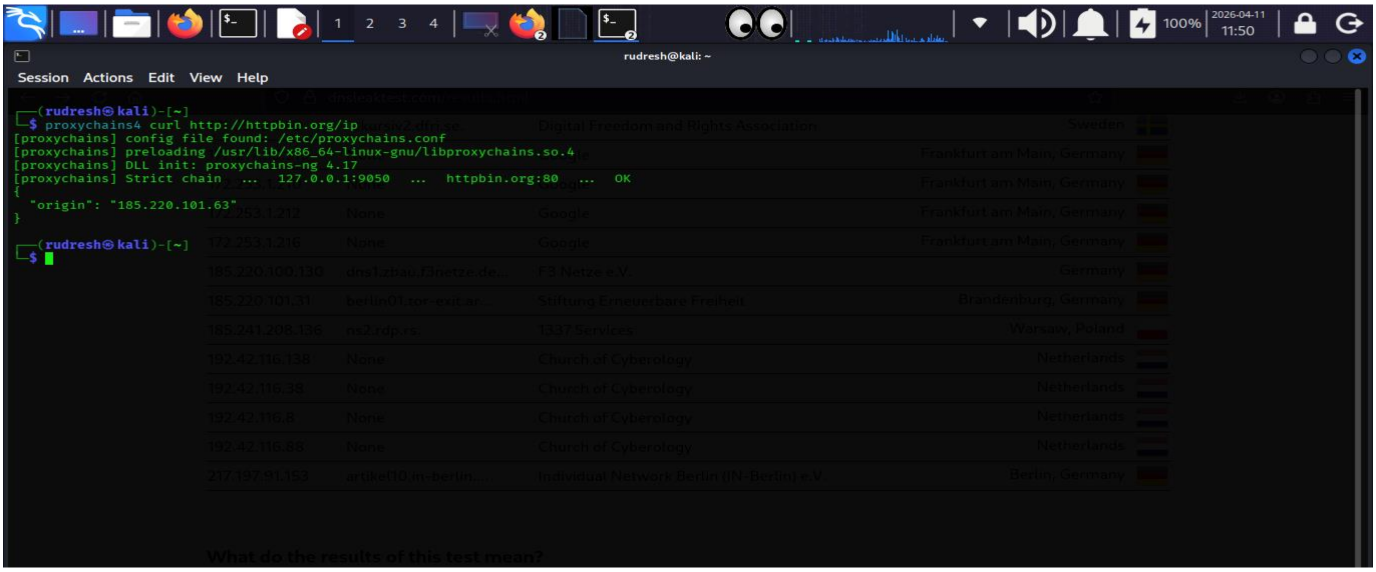
- Set:

```
socks5 127.0.0.1 9050
```

4.7 Testing Anonymous Connection

```
proxychains4 curl http://httpbin.org/ip
```

Result:



4.8 IP Rotation Testing

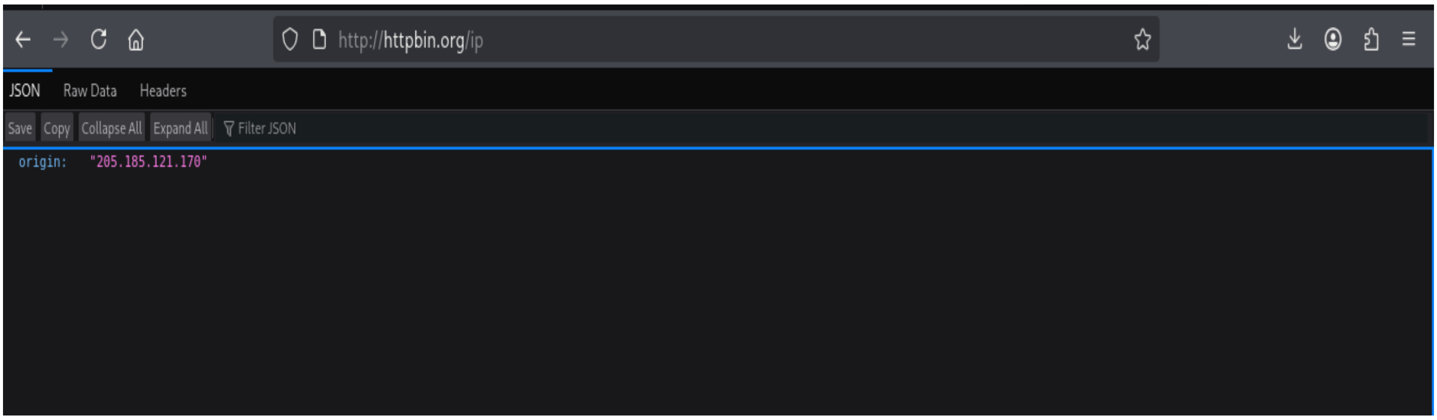
```
sudo service tor@default restart
```

Then:

```
proxychains4 curl http://httpbin.org/ip
```

A new IP address was observed.

Result:



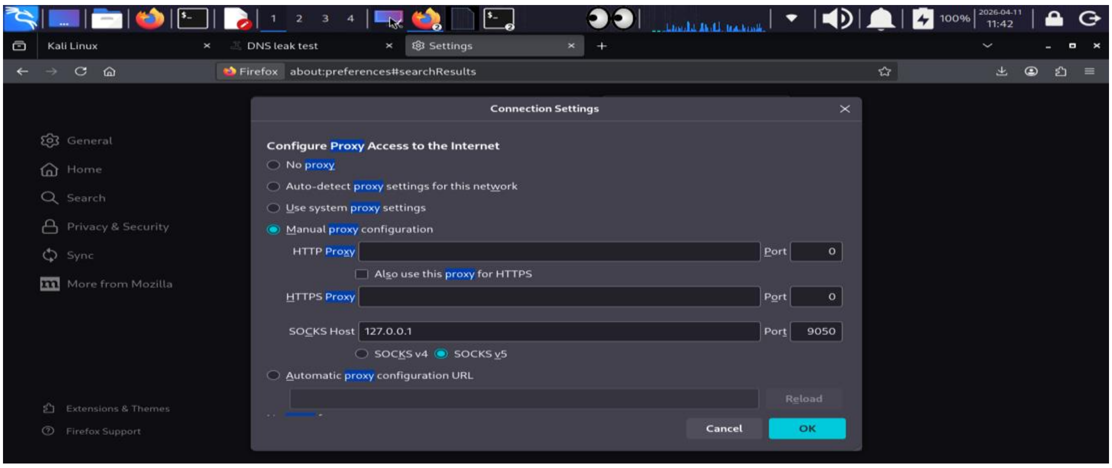
5. Firefox Proxy Configuration

To test browser-level anonymity:

Steps:

- Open Firefox → Settings → Network Settings
- Select Manual Proxy
- Set:
    - SOCKS Host: 127.0.0.1
    - Port: 9050
    - SOCKS v5
    - Enable Proxy DNS

Result:



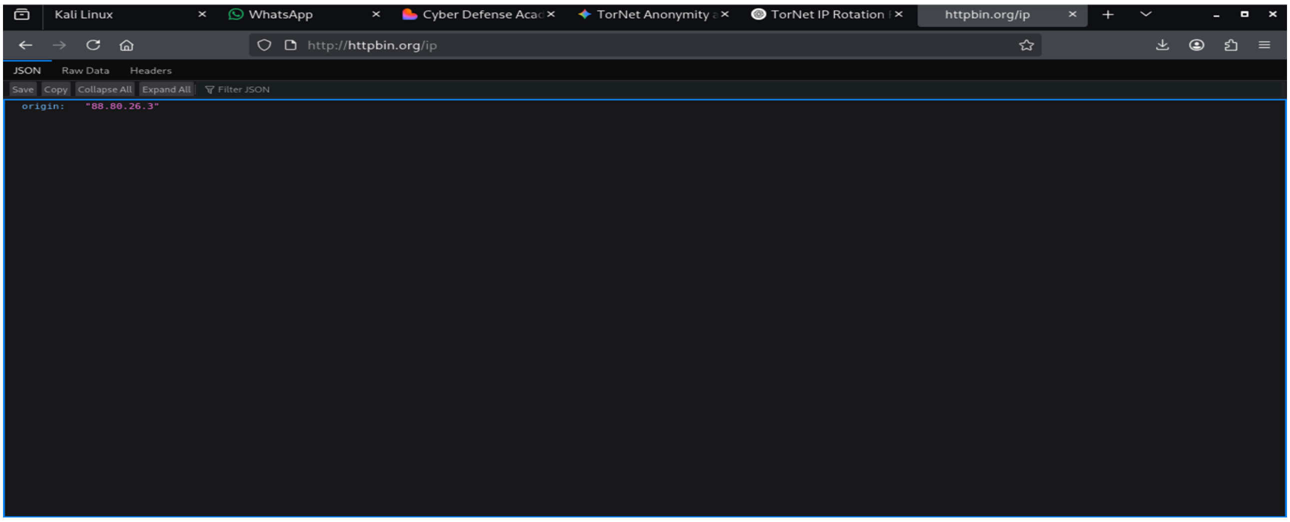
5.1 Browser IP Test

Visited:

- http://httpbin.org/ip

Confirmed Tor IP is visible in browser.

Result:



6. DNS Leak Testing

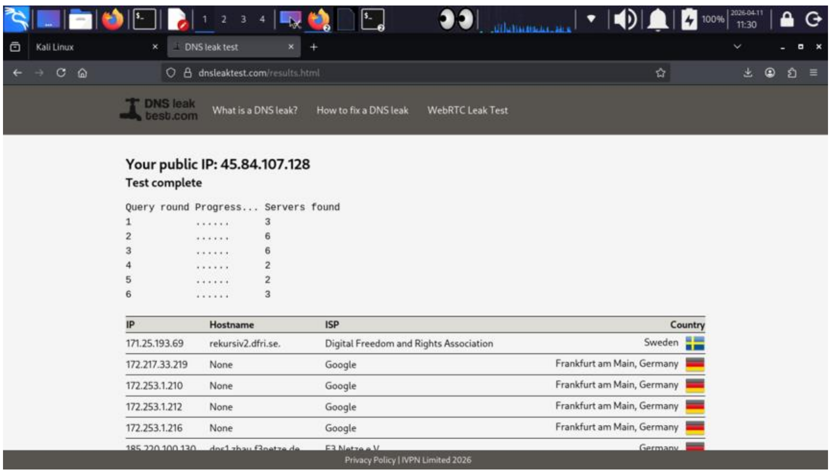
DNS leak testing was performed using:

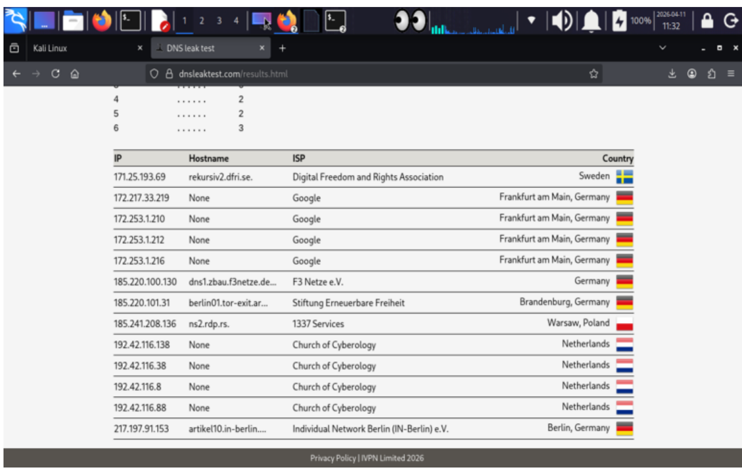
DNSLeakTest

Results:

- Multiple DNS servers detected
- No ISP information visible
- Servers located in different regions

This confirms no DNS leak.





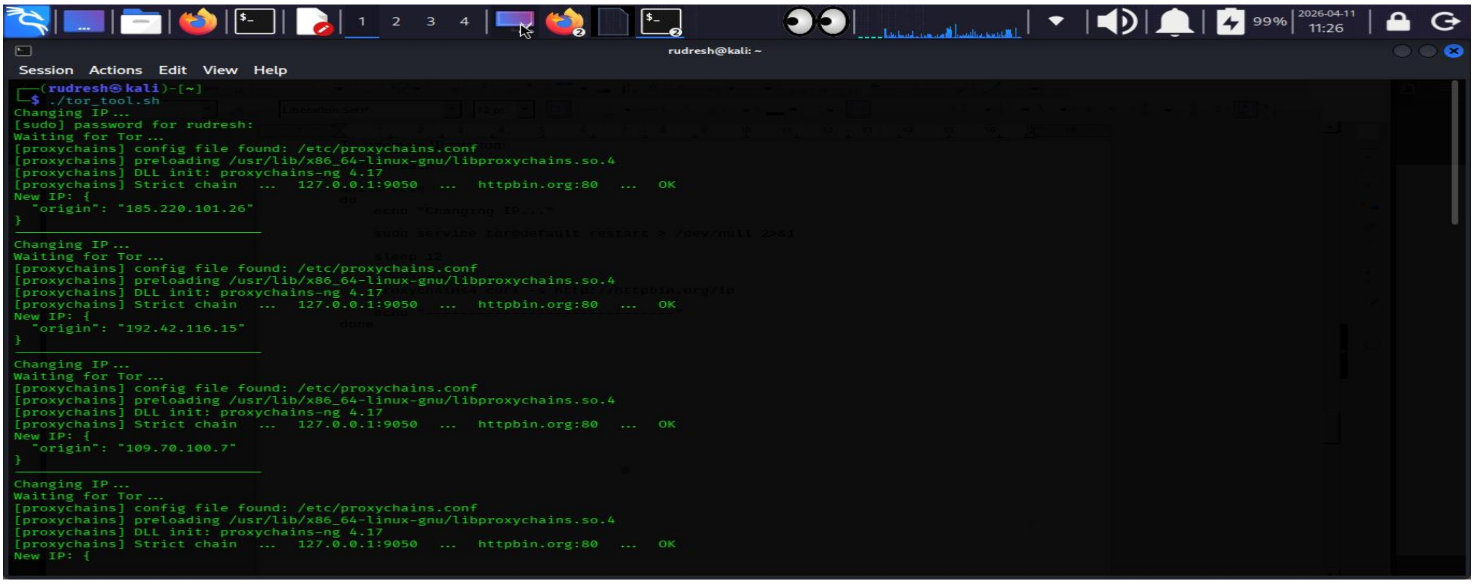
7. Automation Using Bash Script

To automate IP rotation:

```
#!/bin/bash

while true
do
    echo "Changing IP..."

    sudo service tor@default restart > /dev/null 2>&1

    sleep 12

    echo "New IP:"
    proxychains4 curl -s http://httpbin.org/ip

    echo "------------------------------"
done
```

==

Result:



9. Operational Security (OPSEC)

Advantages:

- Hides real IP
- Prevents tracking
- Useful in cybersecurity research

Limitations:

- Slow network
- Exit node risks
- Not 100% anonymous

10. Skills Acquired

- Tor network configuration
- Proxychains usage
- Browser proxy setup
- DNS leak testing
- Bash scripting
- Debugging real issues

11. Conclusion

In this task, I successfully implemented anonymous communication using the Tor network. I verified IP anonymization and DNS protection. I also developed a script for automatic IP rotation, which improved operational security.

This task provided practical knowledge of anonymity techniques used in cybersecurity.

12. Final Outcome

- Tor configured successfully
- Anonymous browsing achieved
- IP rotation verified
- DNS leak prevented
- Automation implemented